

Advanced Linux Troubleshooting Commands Guide

1. CPU Troubleshooting

```
top
htop
uptime
mpstat -P ALL 1
ps -eo pid,ppid,cmd,%cpu --sort=-%cpu | head
pidstat 1
sar -u 1 5
```

2. Memory Troubleshooting

```
free -m
vmstat 1
ps aux --sort=-%mem | head
watch -n 5 ps aux --sort=-%mem | head
cat /proc/meminfo
pmap -x PID
sar -r 1 5
```

3. Disk Usage & I/O Issues

```
df -h
du -sh /* 2>/dev/null
iostat -xm 1
iotop
lsblk
mount | column -t
tune2fs -l /dev/sda1
```

4. Network Troubleshooting

`ip a`

`ip r`

`ss -tulnp`

`netstat -tulnp`

`ping host`

`traceroute host`

`mtr host`

`tcpdump -i eth0`

`ethtool eth0`

5. Process & Service Debugging

`systemctl status service`

`journalctl -xe`

`journalctl -u service`

`strace -p PID`

`lsof -i :PORT`

`lsof | grep deleted`

`kill -SIGTERM PID`

`kill -SIGKILL PID`

6. Log Analysis & Filtering

`tail -f /var/log/syslog`

`less /var/log/messages`

`grep -i ERROR logfile`

`grep -r Exception /var/log`

`awk '{print $1}' file`

`sed -n '1,100p' file`

7. Boot & System Issues

dmesg | less
journalctl -b
journalctl -xb
systemctl list-units --failed
cat /proc/cmdline

8. Security & Access Issues

cat /var/log/auth.log
last
who
w
faillog
passwd -S username

9. Resource Limits & Performance

ulimit -a
top -o %MEM
free -h
sar -q
watch df -h

10. Advanced Debugging Tools

strace command
ltrace command
perf top
perf stat command
htop
vmstat -s

